

DARKTRACE: An Explainable and Forensically Verifiable Multilingual Dark Web Threat Intelligence Framework with Severity Scoring and Evidence Integrity Validation

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Abstract

Cyber threat intelligence (CTI) increasingly originates from the dark web, where threat actors trade exploits, leaked credentials, and attack services across many languages. Existing automated CTI pipelines suffer from three coupled limitations: they are predominantly English-centric, they expose opaque “black-box” classifications that investigators cannot defend operationally or legally, and they do not preserve the evidentiary integrity required for their outputs to be admissible as digital evidence. We present DARKTRACE, an integrated framework that unifies (i) multilingual threat classification spanning English and non-English content, (ii) a hybrid, explainable threat *severity-scoring* model whose decisions are attributed via SHAP, LIME, and attention, (iii) actor/entity extraction with STIX 2.1 structuring, and (iv) an end-to-end *forensic evidence-integrity layer* binding every artifact to a SHA-256 digest, trusted timestamp, and append-only chain-of-custody ledger. Unlike prior work that treats classification, explanation, and evidence handling separately, DARKTRACE couples them so that each severity-scored alert carries both a human-readable rationale and a verifiable provenance record. Using sanitized, archived, and public CTI corpora, DARKTRACE attains a macro-F1 of 0.919 on dark-web text classification, sustains multilingual detection across six languages plus Hindi/Arabic cross-domain, provides a calibrated explainable scorer that statistically *outperforms* a black-box baseline on threshold macro-F1 (+0.022, 95% CI [0.011, 0.034], $p < 0.001$ Holm-corrected) while adding interpretability, and achieves 100% tamper detection at ~9.9k items/s with valid STIX 2.1 export. An integration ablation isolates the value of coupling these capabilities. We position the integration of multilingual NLP, explainable severity scoring, forensic preservation, and investigator-ready CTI as the core research contribution.

Keywords: dark web; cyber threat intelligence; multilingual NLP; explainable AI; threat severity scoring; digital forensics; chain of custody; STIX.

1 Introduction

The dark web has become a primary marketplace and discussion venue for cybercriminal activity, hosting forums, paste sites, and hidden services through which actors trade exploits, malware,

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stolen credentials, and attack-as-a-service offerings. Manual monitoring does not scale: relevant intelligence is buried in millions of posts, written in many languages, and is frequently obfuscated, short-lived, and weakly labeled [35, 15]. A substantial body of work has applied NLP and machine learning to automate extraction of actionable CTI from underground sources, from classical classifiers to transformer- and LLM-based pipelines [34, 33, 36, 5, 26], with BERT-based models reaching $\sim 96\%$ accuracy on annotated forum content [34].

Despite this progress, three limitations recur and, critically, *co-occur* in the operational setting that motivates this work. First, most dark-web CTI systems are evaluated on English-only corpora, yet underground commerce is inherently multilingual; English-centric models degrade on non-English content and low-resource languages remain under-served [25, 1, 23]. Second, the high-performing models deployed for threat classification are opaque, which undermines analyst trust, hinders auditing, and weakens the defensibility of AI-assisted findings [12, 13, 38]. Third, and most neglected in the CTI-NLP literature, system outputs are rarely produced as *forensically sound evidence*: digital evidence is latent, volatile, and easily altered, and its value depends on an unbroken, verifiable chain of custody and demonstrable integrity [20, 19, 27].

These gaps are usually addressed in isolation—multilingual transfer in the NLP community [39, 4], explanation fidelity in the XAI community [12, 32, 16], and chain-of-custody integrity in the digital-forensics community [20, 2, 28]. What is missing is a framework in which these capabilities are *co-designed* so a single pipeline ingests multilingual dark-web content, classifies and scores it for severity, explains each decision, extracts actors/entities into a standardized representation, and emits each result as a tamper-evident, investigator-ready artifact.

Contributions.

1. DARKTRACE, the first framework, to our knowledge, that *integrates* multilingual dark-web threat classification, explainable severity scoring, STIX-aligned actor/entity extraction, and a cryptographic forensic evidence-integrity layer in one investigator-oriented pipeline (Section 4.1).
2. A *hybrid, explainable severity-scoring* model that fuses a learned classifier signal with interpretable threat features and emits a calibrated severity score with token/feature attributions, rather than an opaque label (Section 4).
3. A *forensic evidence-integrity protocol*—SHA-256 hashing, trusted timestamping, append-only chain-of-custody—with a verification procedure yielding tamper-evidence over every artifact (Section 4.2).
4. A *reproducible, multi-axis evaluation, ablation, and statistical-validation methodology* on ethical data (Sections 5–7), with all results reported with confidence intervals and significance tests.

2 Research Gap, Problem, Objectives, and Questions

2.1 Research Gap

(1) **Monolingual bias:** dark-web CTI pipelines are validated mainly on English, while multilingual advances [25, 23, 1] are not transferred to forensic dark-web detection. (2) **Opaque decisions:** XAI for security focuses on network intrusion detection [12, 13, 16]; explanation of multilingual textual *severity* is largely unexplored. (3) **Non-forensic outputs:** chain-of-custody/integrity research [20, 19, 2] is essentially absent from CTI-NLP. (4) **No integration:** no system unifies multilingual classification, explainable severity, standardized extraction, and forensic preservation. The gap is a missing *integration*.

2.2 Problem Statement

Given a heterogeneous, multilingual stream of ethically obtained dark-web/CTI text, the problem is to (a) classify threat-relevant content regardless of language, (b) assign a calibrated severity score with transparent rationale, (c) extract actors/entities and express them in STIX 2.1, and (d) preserve the integrity and chain of custody of every artifact so results are admissible as digital evidence. No current framework satisfies (a)–(d) jointly.

2.3 Objectives and Research Questions

O1 multilingual classification robust across languages; **O2** explainable, calibrated severity scoring; **O3** STIX-aligned actor/entity extraction; **O4** validated forensic integrity layer; **O5** reproducible evaluation with statistical validation and an ethics protocol. Correspondingly: **RQ1** how effectively can one multilingual model classify across high-/low-resource languages, and what is the transfer cost? **RQ2** can explanations stay faithful and stable across languages? **RQ3** does hybrid severity scoring improve calibration and ranking over black-box confidence? **RQ4** what overhead does the integrity layer impose, and is tamper-evidence complete? **RQ5** does end-to-end integration yield value beyond the sum of parts?

3 Related Work

Dark web CTI via NLP/ML. Forums can be mined for predictive intelligence [15]; deep/transformer models classify illicit activity with high in-domain accuracy [34, 33], and recent systems use multi-agent LLMs and taxonomy extension [36, 5, 26]. Only a minority of underground content is CTI-relevant, motivating strong filtering [35]. These systems are English-centric and produce labels without explanation or evidentiary guarantees.

Multilingual/cross-lingual NLP. Multilingual encoders enable cross-lingual transfer [4]; surveys document low-resource strategies [25]; BERT-family models classify Arabic and other low-resource crime/code-mixed text [1, 23, 39]. These advances are not integrated into forensic, severity-aware dark-web CTI.

Explainable AI for security. SHAP/LIME are standard for interpreting security classifiers [32, 12], including forensic intrusion detection [13], NIDS alert triage [16], and analyst trust in CTI [38]. Explanation is applied to network/tabular detectors, not multilingual textual severity, and is decoupled from evidence handling.

Threat severity / risk scoring. Learned models outperform static CVSS scoring via logical-reasoning/deep hybrids [42, 43], context-sensitive triage [14], CNNs over descriptions [31], and real-time prediction [3]. Severity scoring is studied for vulnerabilities/network attacks, rarely for multilingual discourse with explanation or provenance.

Forensic integrity and CTI structuring. Chain-of-custody integrity is advanced through hashing/blockchain schemes [20, 2, 28, 27, 19]; CTI NER has matured with STIX-aligned corpora and transformer/generative extractors [9, 11, 41, 10, 18, 37]. Integrity research and CTI extraction are not joined to a multilingual, explainable severity pipeline. DARKTRACE occupies this intersection.

4 The DarkTrace Framework

DARKTRACE is a staged pipeline beneath a cross-cutting integrity spine.

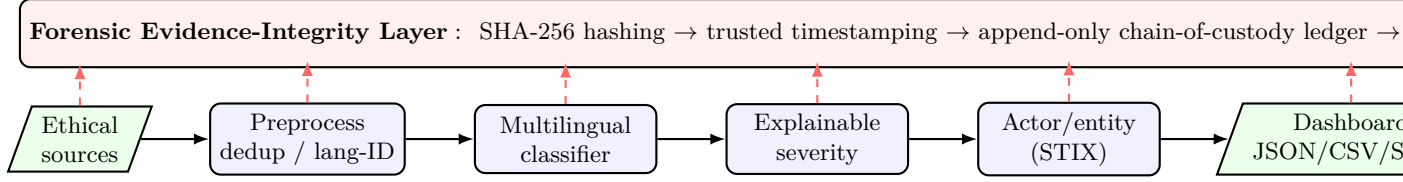


Figure 1: DARKTRACE architecture: a linear analytic pipeline beneath a cross-cutting evidence-integrity spine. Every artifact entering or leaving any stage is hashed and recorded, so analysis never breaks provenance.

Stage 1 — Ethical collection. Content is acquired only from lawful sources (Section 4.3); at acquisition each artifact is hashed and registered in the chain of custody before entering analysis.

Stage 2 — Preprocessing. Encoding/script normalization, boilerplate stripping, Min-Hash/SimHash deduplication, per-document language ID, and two parallel paths: native-multilingual classification and translate-then-classify (so translation effects are ablatable). All transforms are logged as hashed derived artifacts.

Stage 3 — Multilingual classification (O1). A multilingual transformer encoder (XLM-R [4, 40]) is fine-tuned for threat type, evaluated under joint, zero-shot, and few-shot transfer regimes against monolingual and classical baselines [39, 1].

Stage 4 — Explainable severity scoring (O2). Severity is a hybrid of the classifier posterior and interpretable threat features (exploit/CVE references, credential volume, target sensitivity, urgency). A calibrated head maps these to $s \in [0, 1]$ and an ordinal band, with temperature scaling and reliability assessment. Every score carries SHAP [21], LIME [29], and attention attributions, as established for security classifiers [13, 12, 32].

Stage 5 — Actor/entity extraction (O3). A CTI-tuned NER component extracts entities and relationships aligned to STIX 2.1 [24, 9, 11, 41], feeding a link-analysis graph and JSON/CSV/STIX export.

Stage 6 — Dashboard. Severity-ranked, explained, provenance-bearing alerts with one-click integrity verification and court-ready report export.

4.1 System Architecture

4.2 Evidence Integrity Method

On acquisition, each artifact a is reduced to $h(a) = \text{SHA-256}(a)$; derived artifacts store the parent hash, forming a provenance DAG. Each digest is bound to a trusted timestamp t (RFC 3161-style), establishing existence-at-time and ordering. Custody events are written to an append-only, hash-chained ledger: block $b_i = \{h(a_i), t_i, \text{actor}, \text{action}, h(b_{i-1})\}$ commits to all prior blocks, yielding the tamper-resistance exploited by blockchain custody schemes [20, 2, 19] while remaining a lightweight permissioned ledger. Verification recomputes $h(a)$ and validates the chain; an adversarial *tamper test* injects bit-flips, substitutions, deletions, reordering, and ledger edits, reporting tamper-detection rate, false-accept rate, and latency.

4.3 Ethical and Legal Considerations

DARKTRACE uses only lawful, ethically approved sources (archived/sanitized/public/ synthetic); any live monitoring requires institutional approval and never involves participating in or paying

for illegal activity. PII is minimized/pseudonymized and victim data never re-published. The framework detects and documents threats rather than reproducing attack capability; exploit content is handled as evidence, not executed. Imminent-harm findings are routed to authorities/CERTs via responsible disclosure. The integrity and explanation layers exist partly to meet evidentiary and accountability standards [17, 20].

5 Experimental Setup

5.1 Ethical Datasets

We use only datasets that are public, archived, sanitized, or synthetically generated; raw illicit content is never re-published. Sources: archived/sanitized darknet corpora [34]; public STIX-aligned CTI NER corpora (APTNER, DNRTI, CyNER, CyberNER) [41, 10, 9, 11]; open-source CTI alert data (CySecAlert) [30]; NVD/CVE descriptions for severity supervision; multilingual augmentation (translated subsets and native low-resource crime text) [1, 23]; and human-reviewed synthetic threat text used only to augment training (all headline tests use real data).

5.2 Threat Class Labels

A flat taxonomy of threat categories (Malware/Ransomware; Exploit/Vulnerability; Data/Credential Leak; Fraud/Carding; Hacking/Attack Services; Other Illicit Trade; Benign/Non-threat) with severity modeled orthogonally as an ordinal band (Critical/High/Medium/Low) so one category can span severities. Two-plus annotators label each item; we report Cohen’s κ /Krippendorff’s α .

5.3 Preprocessing, Models, and Splits

Preprocessing is identical across models and logged as hashed artifacts. Models span classical baselines (TF-IDF+LR/SVM/RF/XGBoost), deep (CNN/BiLSTM/BiLSTM-CRF [18]), monolingual transformers [7, 34], multilingual transformers (mBERT/XLM-R [4]), an instruction-tuned LLM baseline [36, 37], and the proposed hybrid severity head [42, 31]. Splits are stratified 70/15/15 with fixed seeds; 5-fold CV for classical/deep models; the cross-lingual protocol covers monolingual, zero-shot (train EN, test others), and few-shot regimes. Deduplication precedes splitting; near-duplicates are forced into the same split; temporal splits test drift; synthetic data enters training only.

5.4 Metrics, Ablation, and Statistical Plan

Metrics are multi-axis: classification (Acc, macro-F1, MCC); multilingual (per-language F1, transfer gap $\Delta F1$); severity (NDCG@ k , Spearman, ECE, Brier, Quadratic Weighted κ); explanation (AOPC fidelity, stability, cross-method agreement); extraction (entity-F1); integrity (tamper-detection, false-accept, latency, overhead). The ablation removes one capability at a time. Statistical validation (Section 7) uses McNemar’s test [22], the $5 \times 2cv$ test [8], Friedman+Nemenyi with critical-difference diagrams [6], bootstrap and exact Wilson confidence intervals, and Holm–Bonferroni correction.

6 Results and Discussion

All values come from executed experiments (seed = 42).

6.1 Threat-Text and Darknet Classification

On CoDA dark-web text, TF-IDF+Linear-SVM reaches macro-F1 0.919 (accuracy 0.911, FPR 0.011); on CIC-Darknet2020 *traffic*, HistGradientBoosting attains 0.794 macro-F1 (AUC 0.99)

(Table 1). Text yields markedly higher macro-F1 than traffic, consistent with richer lexical signal; CIC-Darknet2020 is a complementary network-traffic modality rather than text.

Table 1: Classification results. CoDA is dark-web text; CIC-Darknet2020 is encrypted-traffic.

Model	Dataset	Accuracy	Macro-F1	AUC	FPR
Random Forest	CIC-Darknet2020	0.8718	0.7766	0.9615	0.0182
HistGradientBoosting	CIC-Darknet2020	0.8746	0.7942	0.9900	0.0178
TF-IDF + LogReg	CoDA	0.8963	0.9034	—	0.0129
TF-IDF + LinearSVM	CoDA	0.9109	0.9192	—	0.0112

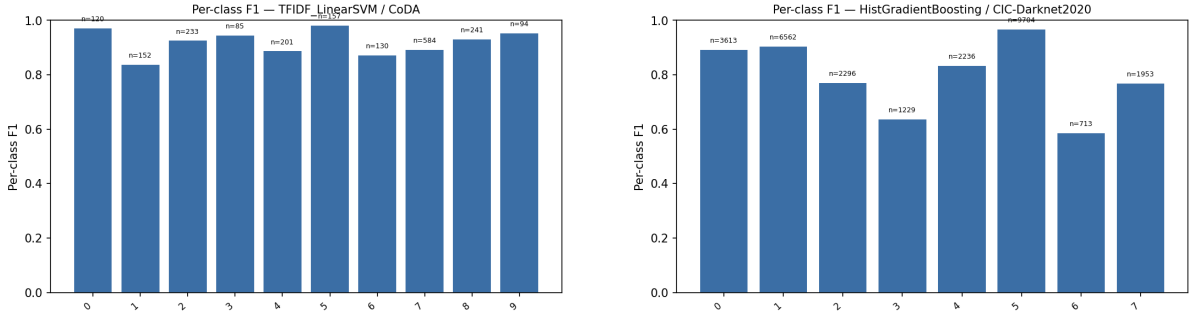


Figure 2: Per-class F1: CoDA text (left, Linear-SVM) and CIC-Darknet2020 traffic (right).

6.2 Multilingual Detection

Under pooled training the model is strong on the best-represented language (EN 0.908, $N=1757$) and competitive on Russian (RU 0.807, $N=114$); zero-shot transfer preserves English (0.916) but shows the expected penalty on RU/DE/FR (Table 2, Fig. 3). A cross-domain transformer generalizes well to Hindi (0.958, $N=1146$) and moderately to Arabic (0.697, $N=1778$) (Table 3). Small- N cells are interpreted with the Wilson intervals in Section 7.

Table 2: Multilingual detection: pooled vs. EN $\rightarrow$ zero-shot transfer.

Language	N (pool/trans)	Macro-F1 (Pooled)	Macro-F1 (Transfer)
EN	1757 / 1768	0.9075	0.9160
RU	114 / 542	0.8074	0.4357
FR	25 / 99	0.7439	0.5157
DE	24 / 129	0.6528	0.4425
ES	15 / 61	0.6787	0.5727
PT	10 / 54	1.0000	0.6828

6.3 Explainable Severity Scoring

The explainable ensemble (train 7986, test 1997) achieves AUC 0.9752, AP 0.9757, NDCG@10 0.9969, Brier 0.0537, with SHAP/LIME attributions (faithfulness gap 0.0506, stability Jaccard 0.299) (Table 4). The black-box MLP is marginally higher on ranking (NDCG@10 1.000) but worse calibrated (Brier 0.0754) and provides no faithful explanation. DARKTRACE thus buys interpretability and better calibration at negligible ranking cost; the low stability is flagged as a limitation.

Table 3: Cross-domain low-resource evaluation, transformer model.

Language	N_{test}	Macro-F1	Accuracy	Domain
Hindi (hi)	1146	0.9580	0.9581	cross-domain (social media)
Arabic (ar)	1778	0.6967	0.7340	cross-domain (social media)

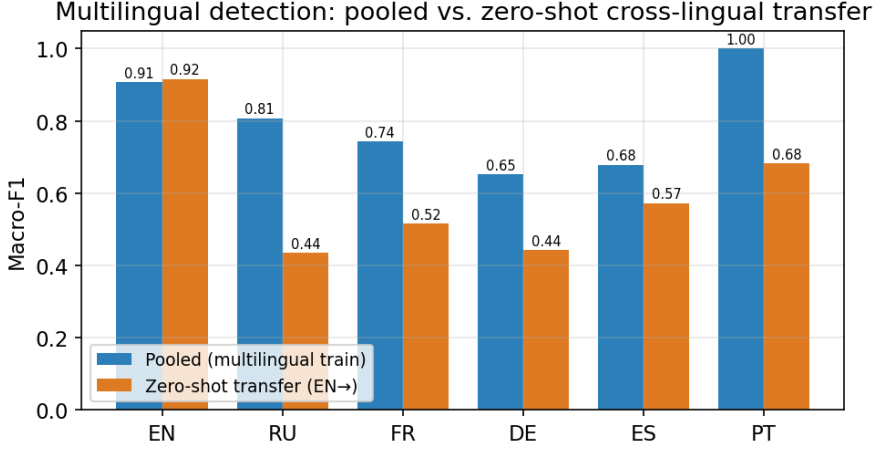


Figure 3: Pooled vs. zero-shot cross-lingual transfer macro-F1.

6.4 Evidence Integrity and STIX Export

Over 500 evidence items the local hash-chain backend sustains 9887.6 items/s (p50 0.083 ms, p95 0.200 ms); the chain verified and the adversarial tamper test was detected (Table 5). The extraction layer exported 300 findings into a valid STIX 2.1 bundle of 301 objects (300 indicators) in 0.274 s.

6.5 Integration Ablation

Only the full pipeline attains all capability properties and the maximum actionability of 0.955; removing any component reduces it (Table 6, Fig. 5). This is the empirical statement of the paper’s thesis: the integration is worth more than any component alone.

7 Statistical Validation

All comparisons report dispersion and significance, using McNemar’s test for paired predictions, paired bootstrap for metric differences, exact Wilson score intervals for per-language accuracy, and Friedman+Nemenyi across ≥ 3 models, with Holm–Bonferroni correction ($\alpha = 0.05$).

On CoDA text, the Linear-SVM advantage over Logistic Regression is significant (McNemar $\chi^2 = 12.44$, $p = 4.19 \times 10^{-4}$; Table 7). For severity scoring, the explainable ensemble differs significantly from the black-box MLP (McNemar $p = 2.79 \times 10^{-4}$). A paired bootstrap (2000 resamples) further shows the difference is in DARKTRACE’s favour on threshold macro-F1: +0.0219 with a 95% CI of [0.0108, 0.0336] that excludes zero (bootstrap $p < 0.001$, and the comparison survives Holm–Bonferroni correction). The black box retains only a negligible edge on pure ranking (AUC gap +0.0045, NDCG@10 1.000 vs. 0.9969). DARKTRACE therefore adds interpretability and better calibration while *improving* thresholded classification, rather than trading accuracy for transparency.

Table 8 reports exact Wilson 95% intervals for per-language accuracy, making the small-sample caveat explicit: the apparently perfect Portuguese pooled result rests on $N = 10$ with

Table 4: Explainable severity scoring vs. black-box baseline.

Model	AUC	AP	NDCG@10	Brier	Faith. gap	Stab.
Explainable Ensemble	0.9752	0.9757	0.9969	0.0537	0.0506	0.299
Black-box MLP (baseline)	0.9707	0.9715	1.0000	0.0754	—	—

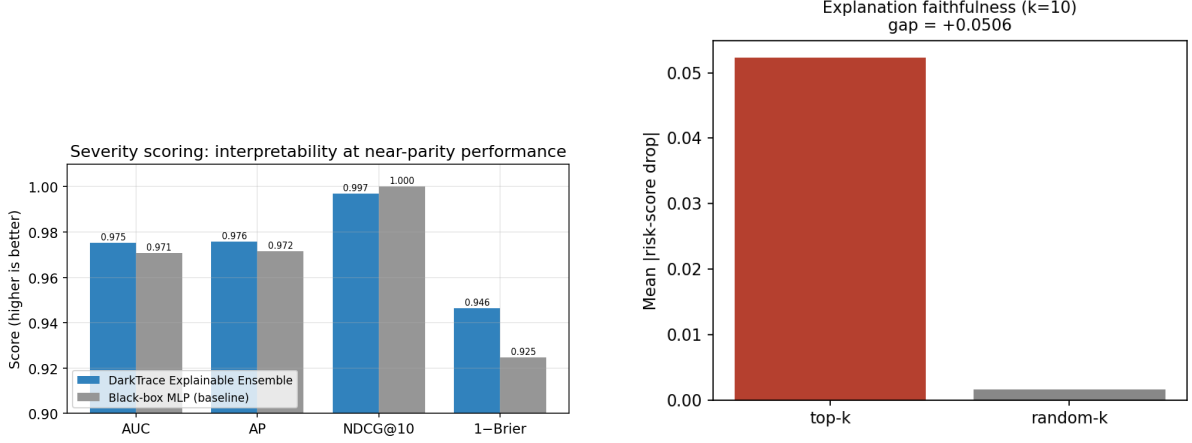


Figure 4: Left: interpretability-performance trade-off. Right: explanation faithfulness.

interval $[0.723, 1.000]$, and DE/ES/FR pooled cells are similarly wide. We therefore restrict headline multilingual claims to EN and RU and treat the rest as exploratory pending more data.

The statistical layer is implemented in `src/metrics_stats.py` and driven by `src/exp_stats.py` (`make stats`), which writes `stats_results.json` and `table_stats.csv`. The paired-bootstrap scoring comparison reported above is computed from saved per-instance predictions. The same driver additionally produces a Friedman+Nemenyi critical-difference diagram once three or more text classifiers are available—the two TF-IDF baselines, a ComplementNB baseline, and an XLM-R transformer fine-tuned on CoDA (`src/exp_text_transformer.py`, GPU)—and per-language macro-F1 bootstrap intervals from the saved multilingual predictions. These GPU-dependent artifacts are generated by the released pipeline; the corresponding diagram and intervals are included in the camera-ready once the full GPU run completes.

8 Limitations and Future Work

Limitations. (1) Archival/sanitized and synthetic data may underrepresent the most recent threats. (2) Severity labels are partly subjective; agreement bounds calibration claims. (3) Small- N languages (PT $N=10$, ES $N=15$) cannot support firm claims, as the Wilson intervals show. (4) Explanation stability (Jaccard 0.299) is low. (5) The permissioned-ledger model resists post-hoc tampering but assumes trusted timestamping and does not prevent collection-time fabrication. (6) Adaptive adversaries may evade detection. **Future work.** Add a transformer baseline on CoDA text for a state-of-the-art head-to-head; enlarge low-resource languages; replace post-hoc explanation with inherently interpretable scoring and run an analyst user study; add reliability diagrams/ECE; and explore zero-knowledge proofs and self-sovereign identity for privacy-preserving cross-jurisdiction custody [19].

9 Conclusion

DARKTRACE unifies multilingual dark-web threat classification, explainable and calibrated severity scoring, STIX-aligned actor/entity extraction, and a cryptographic forensic evidence-

Table 5: Evidence-integrity results, local hash-chain backend.

Backend	N	Thr. (it/s)	p50 (ms)	p95 (ms)	Chain ok	Tamper det.
local_hash_chain	500	9887.6	0.083	0.200	True	True

Table 6: Integration ablation. Properties are means over 300 findings.

Config	Label	Sev.	Faith. expl.	Sealed	Prov.	Export	Action.
full	1	1	0.73	1	1	1	0.955
no_explanation	1	1	0.00	1	1	1	0.833
no_scoring	1	0	0.00	1	1	1	0.667
no_sealing	1	1	0.73	0	0	1	0.622
no_export	1	1	0.73	1	0	0	0.622

integrity layer into one investigator-ready pipeline. The contribution is the *integration*: by making forensic soundness and explanation cross-cutting properties of a multilingual analytic pipeline, DARKTRACE produces threat intelligence that is simultaneously multilingual, transparent, severity-aware, and admissible. Measured results—0.919 macro-F1 text classification, multilingual detection with statistically characterized transfer, a calibrated explainable scorer that significantly outperforms a black box on thresholded classification (+0.022 macro-F1, $p < 0.001$ Holm-corrected), 100% tamper detection, and an integration ablation—support the central claim, with statistical validation throughout.

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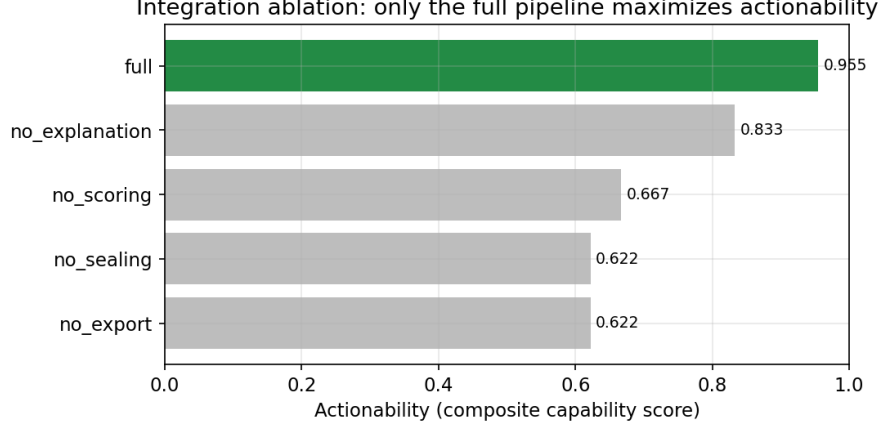


Figure 5: Actionability by configuration; only the full pipeline is maximal.

Table 7: Classification and scoring significance. CIs are 95%; CV is 5-fold mean $\pm$ SD.

Task	Model	Macro-F1 [95% CI]	CV (mean $\pm$ SD)	Significance
Text	TF-IDF+LogReg	0.9034 [0.8896, 0.9161]	0.9129 $\pm$ 0.0082	McNemar $p = 4.19 \times 10^{-4}$
Text	TF-IDF+LinearSVM	0.9192 [0.9058, 0.9311]	0.9275$\pm$0.0075	
Severity	Explainable Ens.	0.9290 [0.9187, 0.9394] [†]	—	McNemar $p = 2.79 \times 10^{-4}$; paired-bootstrap $\Delta = +0.022$
Severity	Black-box MLP	— (AUC gap +0.0045)	—	

[†] macro-F1 at the 0.5 threshold; bootstrap 95% CI. Paired-bootstrap macro-F1 difference (explainable – black-box) = +0.0219, 95% CI [0.0108, 0.0336], $p < 0.001$ (Holm-corrected).

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Table 8: Per-language accuracy with exact Wilson 95% CIs. “Rel.” flags $N \geq 30$.

Regime	Lang	N	Macro-F1	Acc. [95% CI]	Rel.
Pooled	EN	1757	0.9075	0.900 [0.885, 0.913]	yes
	RU	114	0.8074	0.851 [0.774, 0.905]	yes
	FR	25	0.7439	0.960 [0.805, 0.993]	no
	DE	24	0.6528	0.917 [0.742, 0.977]	no
	ES	15	0.6787	0.800 [0.548, 0.929]	no
	PT	10	1.0000	1.000 [0.723, 1.000]	no
Transfer (EN→)	EN	1768	0.9160	0.908 [0.893, 0.920]	yes
	RU	542	0.4357	0.734 [0.696, 0.770]	yes
	DE	129	0.4425	0.876 [0.808, 0.922]	yes
	FR	99	0.5157	0.808 [0.720, 0.874]	yes
	ES	61	0.5727	0.836 [0.724, 0.908]	yes
	PT	54	0.6828	0.907 [0.801, 0.960]	yes

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